Chapter 8

REGRESSION ANALYSIS

8.1 INTRODUCTION

The analysis of coefficient of correlation between the two variables which we have studied in the previous chapter, shows the extent and degree of relationship between the two variables. However, if the two variables are related, it means that when one changes by a certain amount, the other also changes on an average by a certain amount. The statistical technique that expresses a functional relationship between the two variables in the form of an equation to estimate (or predict) the value of a variable, based on the given value of another variable is called **regression analysis.**

Through regression equations we can quantify the relationship between two variables and we can predict the average value of one variable corresponding to a specific value of the other.

8.2 MEANING AND DEFINITIONS

The dictionary meaning of the word *regression is 'stepping back'* or *'going back'*. The term *'regression'* was first used by the English biometrician, **Sir Francis Galton** in the nineteenth century, when he was studying the heights of about one thousand fathers and sons. His study revealed :

Tall fathers tend to have tall sons and short fathers short sons.

The average height of the sons of a group of tall fathers is less than that of the fathers, while the average height of the sons of a group of short fathers is greater than that of the fathers.

The line describing the tendency to regress or going back was called by **Galton** a *'Regression Line'*. But now a days many statisticians prefer to use the term *'estimating line'* instead of *'regression line'*.

Today the word regression is used in statistics has a much wider scope without any reference to biometry. It is used in almost all the sciences – social, natural or physical. In economics it is the basis of estimating the relationship among economic variables which helps a lot in economic theory.

The variable whose value is to be estimated is called *dependent variable* and the variable whose value is used to estimate this value is called *independent variable*. Thus in regression, we can estimate the unknown value of one (dependent) variable from known value of the other (independent) variable.

The independent variable is also called 'Regressor' or 'Explaining Variable' and the dependent variable is called the 'Regressed' or 'Explained Variable'.

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Some definitions of Regression are :

1. "Regression is the measure of the average relationship between two or more variables in terms of the original units of the data." –Morris M. Blair

2. "One of the most frequently used techniques in economics and business research to find a relation between two or more variables that are related casually, is regression analysis." –Taro Yamane

3. "Regression analysis measures the nature and extent of the relation between two or more variables, thus enables us to make prediction." –Hirsh

4. "Regression analysis attempts to establish the nature of relationship between variables.... and thereby provide a mechanism for prediction or forecasting." –Ya Lun Chou

8.3 UTILITY OF REGRESSION ANALYSIS

Regression analysis is a branch of statistical theory that is widely used in almost all the scientific disciplines.

The main uses of regression analysis are as follows :

1. **Nature of Relationship.** Regression analysis is useful in establishing the nature of relationship between two variables.
2. **Estimation of Relationship.** Regression analysis is used to estimate the strength of the relationship between two variables.
3. **Prediction.** Regression analysis provides a functional relationship between two or more related variables, with the help of which we can easily estimate or predict the unknown value of one variable from the known value of another variable.
4. **Useful in Economics.** Regression analysis is widely used in business and economic activities. Businessmen are interested in predicting future production, investment, consumption, prices, profits, sales etc. and success of business depends very much on the degree of accuracy in the predictions.
5. **Coefficient of Correlation.** With the help of regression studies, we can also calculate the value of coefficient of correlation *i.e.* $r = \sqrt{b_{xy} \times b_{yx}}$.
6. **Cause and effect Relationship.** Regression analysis provides a valuable tool for measuring and estimating the cause and effect relationship among the variables which helps in economic and business research.
7. **Calculation of Coefficient of Determination.** Regression coefficients are generally used in calculating coefficient of correlation $r = \sqrt{b_{xy} \times b_{yx}}$. And square of coefficient of correlation (r) is called the *coefficient of determination* which measure the effect of the independent variable on the dependent variable. The higher the value of $r^2 (r^2 = b_{xy} \times b_{yx})$, the better are regression lines and more useful are the regression equations for estimation and prediction.
8. **Other Fields.** Besides economics, the regression analysis is highly used in other natural, physical and social sciences, e.g. it estimates birth rate, death rate, tax rate etc.
9. **Calculation of Errors.** Regression analysis is helpful in estimating the errors made through regression lines.

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8.4 LIMITATIONS/ASSUMPTIONS OF REGRESSION ANALYSIS

Despite the above utilities, the regression analysis, too has various limitations. The following are some of the limitations of regression analysis :

1. **Assumption of Linear Relationship**. Regression analysis is based on the assumption of linear relationship between related variables. But practically, it is not possible all the times. As in the field of social sciences, non-linear or curvilinear relationship are commonly exist.
2. **Cause and effect Relationship.** In calculating regression equations, the cause and effect relationship between the variables is assumed to be static (unchanged) e.g. at one time, production may be increased due to increase in demand, whereas at another time demand may be increased due to abundance of production. Thus this relationship lead to misleading results sometimes.
3. **Lengthy.** This process involves lengthy and complicated calculations which consumes lot of time and financial resources.
4. **Qualitative Data.** Regression analysis cannot be used in case of qualitative data, e.g. beauty, honesty, intelligency etc. as we have done in correlation analysis.

8.5 TYPES OF REGRESSION ANALYSIS

There are various types of regression analysis as given below :

1. Simple, Multiple and Total Regression

In **simple regression** analysis, we study relationships between only two variables at a time, in which one variable is dependent and other one is independent e.g. relationship between income and expenditure is concerned with simple regression.

In **multiple regression** analysis, we study relationships between more than two variables at a time e.g. study of growth in production of rice in relation to rainfall, irrigation, germicides etc. is concerned with multiple regression.

In **partial regression** analysis, we have more than two variables, but we study, the relationship between only two variables and keeping other variables as constant to excludes the influence of these variables.

2. Linear and Non-Linear Regression

In **linear regression,** a unit change in the value of the independent variable results in a constant or proportionate change in the value of dependent variable. In linear regression, if we plot the data on graph paper it give rise to a straight line and the linear relationship between the variables X and Y are shown as a first degree equation in X and Y (*i.e.* $Y = a + bX$ or $X = a + bY$).

In **non-linear regression,** one variable varies with other variable in a changing ratio and the graph plot for non-linear regression form a curve. That is why, non-linear regression is also called as **curvilinear regression.**

8.6 REGRESSION LINES

The regression line shows the average relationship between the two variables. It is also known as the **Line of Best Fit** or an *estimating line* as it gives the average estimated value of the dependent variable (Y) for any given value of independent variable (X).

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According to **J.R. Stockton,** "The device used for estimating the value of one variable from the value of the other consists of a line through the points, drawn in such a manner as to represent the average relationship between the two variables. Such a line is called the *line of regression.*"

If there are two variables X and Y, then there must be two regression lines related to them, which are as follows :

1. **Regression line of X on Y :** The regression line of X on Y gives the best estimate for the value of X, when the value of Y is given.
2. **Regression line of Y on X :** The regression line of Y on X gives the best estimate for the value of Y, when the value of X is given.

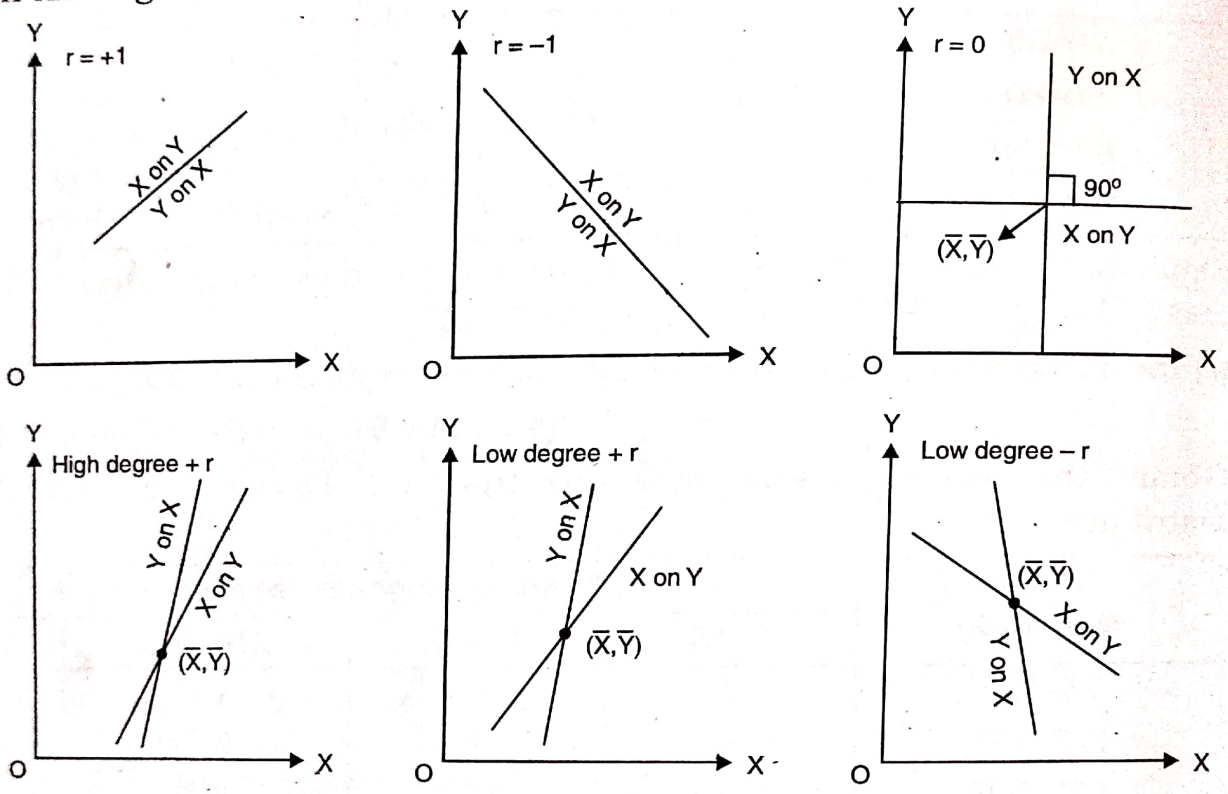
The above two lines are not reversible because the basis and assumptions for deriving these lines are quite different.

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8.11 CORRELATION AND REGRESSION

Correlation and Regression analysis are two important statistical techniques to study the relationship among two or more variables. Coefficient of correlation measure the degree of covariance between the two variables whereas in regression, we study the nature of the relationship between the two variables so that we may get the value of one variable on the basis of value of other.

For better understanding, the following diagrams show the relationship between correlation and regression.



Note. (1) The point where both the regression lines meet is $(\bar{X}, \bar{Y})$.

(2) The higher the value of r, closer will be the regression lines to each other and vice-versa.

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8.13 DIFFERENCE BETWEEN CORRELATION AND REGRESSION

Following are the main differences between correlation and regression :

(1) **Definition.** Correlation is concerned with the measurement of strength and intensity of relationship between the variables, whereas regression is concerned with prediction of the value of dependent variable in relation to a known value of the independent variable.

(2) **Relationship.** Correlation finds out the degree of relationship between two variables and not cause and effect relationship of the variables. But regression indicates the cause and effect relationship between the variables and establishes a functional relationship.

(3) **Mutual Dependence of Variables.** Correlation is a test of interdependence between the two variables, but regression analysis gives a mathematical formula to determine value of the dependent variable with respect to a value of independent variable.

(4) **Units.** The coefficient of correlation is a relative measure and independent of unit of measurement, whereas regression is not. Regression coefficient gives an absolute figure.

(5) **Scope of Application.** Correlation has limited applications, because it is confined only to linear relationship between the variables. But regression has wider applications, as it studies linear and non-linear relationship between the variables.

(6) **Prediction.** Correlation does not help in making predictions whereas regression enable us to make predictions.

(7) **Non-Sense Correlation.** In correlation, there may be chance or spurious or non-sense correlation between the two variables e.g. correlation between prices of shirts and prices of sugar is non-sense correlation. But in regression analysis, there is nothing like non-sense correlation.

(8) **Variables.** In correlation both the variables are considered alike and do not have any differentiation, but in regression analysis variables are differentiated as dependent and independent variables.

(9) **Mathematical Treatment.** Correlation is not very useful for further mathematical treatments. But regression is widely used for further mathematical treatments.

(10) **Symmetric.** Correlation is symmetric in formation *i.e.* $r_{xy} = r_{yx}$. It means there is no difference whether we find relationship between X and Y or Y and X. Whereas Regression coefficients b_{xy} and b_{yx} are not symmetric i.e. $b_{xy} \neq b_{yx}$. It means there is difference between regression coefficient X on Y and regression coefficient Y on X.

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(11) **Origin and Scale.** Correlation is independent of both shift of origin and change of scale. But regression coefficients are only independent of shift of origin, but not of change of scale.

(12) **Measure :** Regression analysis is an **absolute measure** showing a change in the value of Y or X for a corresponding change in the value of X or Y, whereas correlation coefficient is **a relative measure** of linear relationship of X and Y.

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